Machine Learning - Graduate Level Assessment 21 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following can only be used when training data are linearly separable?
   (a) Linear hard-margin SVM.
   (b) Linear Logistic Regression.
   (c) Linear Soft margin SVM.
   (d) The centroid method.

2. Statement 1| Industrial-scale neural networks are normally trained on CPUs, not GPUs. Statement 2| The ResNet-50 model has over 1 billion parameters.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True

3. Statement 1| As of 2020, some models attain greater than 98% accuracy on CIFAR-10. Statement 2| The original ResNets were not optimized with the Adam optimizer.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True

4. Which of the following are the spatial clustering algorithms?
   (a) Partitioning based clustering
   (b) K-means clustering
   (c) Grid based clustering
   (d) All of the above

5. Computational complexity of Gradient descent is,
   (a) linear in D
   (b) linear in N
   (c) polynomial in D
   (d) dependent on the number of iterations

True / False (10 marks)

Answer True or False

6. True or False: Stuart Russell is widely recognized for his work on existential risks posed by artificial intelligence.
7. True or False: Predicting the amount of rainfall in a region based on various cues is a supervised learning problem.
8. True or False: Anomaly detection is another term commonly used for out-of-distribution detection in machine learning contexts.
9. True or False: The use of sampling with replacement in bagging inherently prevents overfitting by itself without considering classifier strength.
10. True or False: ResNeXt models typically utilize ReLU activation functions rather than tanh activation functions.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the rank of the matrix A = [[1, 1, 1], [1, 1, 1], [1, 1, 1]]. Explain the implications of this rank in the context of linear transformations and its relevance in machine learning.
12. Discuss the relationship between PCA and spectral clustering regarding their eigendecomposition processes, and analyze the distinction between classification and regression in the context of logistic and linear regression.

End of Paper